

AI x CITY CLIMATE ACTION HACKATHON 2026

FloodFact AI

Pilot Plan

Phase 1 — Mukuru, Nairobi (kwa Reuben · kwa Njenga · Viwandani)

Prepared September 2026 · Repository: github.com/VinceOdundo/FloodfactAI

Companion documents: [FloodFact-AI-Concept.pdf](#) · [FloodFact-AI-Demonstration.pdf](#)

1. Executive Summary

FloodFact AI is a community-centered flood early-warning and WhatsApp-rumour-verification system built for Nairobi's informal settlements. This Pilot Plan sets out the 4-week Phase 1 pilot in Mukuru, Nairobi, exactly as scoped in the FloodFact AI Requirements Specification (§11, Pilot Scope), and shows how the system already built for this submission — real database schema, a deterministic and unit-tested risk-scoring core, and a working admin, ambassador, and public product — satisfies each of the Requirements Specification's 12 Phase 1 MVP acceptance criteria (§18).

The system is code-complete and runs today in an honestly-labeled sandbox/demo mode with zero paid accounts (see the Demonstration submission). Turning it on for a real pilot is a deployment and partnership-confirmation exercise, not a rewrite: every external dependency already has a working adapter, and docs/SETUP.md walks through provisioning each one.

1.1 Pilot Scope

Item	Pilot definition
Area	Mukuru: kwa Reuben, kwa Njenga, Viwandani
Duration	4-week demo · September–October 2026
Target	500 households, Phase 1
Ambassadors	10 trained youth per ward (30 total)
Channels	WhatsApp + SMS + 3 community notice boards
Data	ArcGIS Living Atlas + Kenya Meteorological Department (KMD)
Trigger	Short rains season, October onset

Source: FloodFact AI Requirements Specification §11 — Pilot Scope.

1.2 Success Targets

Target	Tracked via
80% of households receive alerts within 30 minutes	pilot_metrics.alerts_within_30min_pct — live SQL view, Admin → Pilot Metrics
50% reduction in rumour spread in pilot zones	pilot_metrics.false_information_rate_pct trend + ambassador household spot-checks
100% of ambassadors trained in AI tool use	pilot_metrics.ambassadors_trained_pct — driven by ambassadors.training_status

Source: FloodFact AI Requirements Specification §4 — Goals (Pilot impact).

2. Activities & Timeline

Four weeks, timed to the short-rains onset named in the Requirements Specification's pilot trigger. Every week's exit criteria are checked against real system state — the escalations queue, data-source health panel, and pilot_metrics view — not a status update typed by hand.

Week 1 — Setup & Training

- **Deploy the production Supabase project**, apply the 6 Supabase migrations, and seed the 3 pilot areas — following docs/SETUP.md.
- **Provision live credentials:** WhatsApp, Africa's Talking, Anthropic, Voyage AI — and confirm ArcGIS Living Atlas / KMD feed access (open item, §19).
- **Recruit and train 30 youth ambassadors (10 per ward)** on the ambassador PWA, ground-truth logging, and escalation handling.
- **Confirm the Mukuru gazetteer and safe-zone locations** with local ambassador knowledge, and install the 3 physical community notice boards.

Week 2 — Soft Launch (Shadow Mode)

- **Open WhatsApp, SMS, and web intake** to a limited set of households per ward.
- **Admin team reviews every classification in shadow** before it reaches the public alerts feed, to validate risk-engine calibration against local ground truth.
- **Daily stand-up** reviewing the escalations queue and the data-source health panel.
- **Iterate on** gazetteer coverage and alert wording based on ambassador and household feedback.

Week 3 — Full Operation

- **Full 500-household rollout** across all three wards; shadow review is lifted.
- **Verified and elevated-risk classifications publish directly; the system still routes** insufficient_evidence and high-impact cases to a human, per acceptance criterion 12.
- **Daily tracking** of the three Requirements-Specification success targets via the live dashboard.
- **Weekly community notice-board refresh** as new reports and ambassador ground-truth confirm or correct earlier claims.

Week 4 — Evaluation & Handoff

- **Compute final pilot_metrics** across all three wards from recorded reports, classifications, and ambassador_observations — acceptance criterion 10.
- **Run an ambassador debrief** and a household spot-check survey, as the direct proxy for the rumour-reduction target.
- **Compile a findings report** against every Requirements-Specification acceptance criterion and goal — see §6 of this plan.
- **Present the roadmap** for Phase 2 (additional hazards) and Phase 3 (national scale) — see §8.

3. Team Structure & Roles

3.1 Pilot Delivery Team

Lean by design — a startup team, not a large program office. Ambassador redundancy (10 per ward, not 1) is deliberate: it is the mitigation for the single highest-likelihood pilot risk (see §7).

Role	Count	Responsibility
Pilot Lead	1	Overall delivery owner; partner and city liaison; final escalation point.
Engineering / Platform Lead	1	Owns deployment, credential provisioning, data-source health, and fixes.
Ward Ambassador Coordinator	3 (1/ward)	Manages and schedules that ward's 10 ambassadors; runs daily check-ins.
Youth Ambassador	30 (10/ward)	Ground-truth verification, community trust layer, notice-board upkeep.
M&E / Data Analyst	1	Owns pilot_metrics reporting and the acceptance-criteria evaluation report.
Partnerships & Data Liaison	1	Manages ArcGIS Living Atlas / KMD relationships and messaging-provider accounts.

3.2 System User Roles

The roles the running product itself is built for, distinct from the delivery team above.

Role	Needs / responsibilities
Resident	Submit a flood report, warning, or suspicious message; receive verified alerts and recommended actions.
Youth Ambassador	Receive community reports, support ground validation, explain results locally, distribute trusted information.
System / Operations Admin	Monitor incoming reports, verification status, alerts, data-source health, audit logs, pilot metrics.
Data / Climate Source	Provide machine-readable or operator-accessible weather, rainfall, flood-risk, river, and historical information.
Partner / Emergency Stakeholder	Receive or review verified alerts and coordinate response where appropriate.

Source: FloodFact AI Requirements Specification §5 — Users and Roles.

4. Data & Partner Dependencies

Every row below already has a working code adapter. “Status for pilot” states plainly what is real today versus what needs a pilot-time decision — including the two data sources the Requirements Specification itself flags as still to be confirmed (§19).

Source	Role in the system	Status for pilot
Open-Meteo	Rainfall, forecast, ERA5 reanalysis evidence	Live today — free, keyless public API
Esri ArcGIS Living Atlas	Flood-risk zone evidence	Named in pilot scope; exact layer/access to be confirmed (open item, §19)
Kenya Meteorological Department (KMD)	Regional advisory / rainfall corroboration	Named in pilot scope; data-sharing arrangement to be confirmed (§19)
Meta WhatsApp Cloud API	Resident report intake + alert delivery	Free test number ready; production verification needed for full reach
Africa's Talking	SMS report intake + alert delivery	Kenyan SMS gateway; sandbox proven, pilot account needed
Anthropic Claude	Structured claim extraction + rationale text (never the verdict)	Adapter built; pilot API key needed
Voyage AI + pgvector	Rumor-pattern similarity matching	Adapter built; pilot API key needed
Supabase	Postgres + PostGIS + pgvector + Auth + Realtime, RLS on every table	Migrations validated; pilot project needs provisioning

5. Acceptance Criteria Traceability

All 12 Phase 1 MVP acceptance criteria from the Requirements Specification, mapped to the real code and schema that already satisfy each one.

#	Requirement	How the system satisfies it
1	A resident can submit a WhatsApp or SMS report and the system creates a trackable case.	Webhook routes (whatsapp, africastalking) + the public /api/reports route all write a validated reports row.
2	A report can be associated with a pilot area where location is available.	lib/core/geocode.ts matches claimed location text against the Mukuru gazetteer and resolves pilot_area_id via PostGIS.
3	The verification workflow can retrieve and record evidence from configured data sources.	lib/pipeline/classify.ts fetches rainfall, flood-risk, historical, corroboration, and ambassador signals in parallel; every fetch — success, failure, or sandbox — becomes an evidence_items row.
4	The system produces one of the three defined classifications.	lib/core/risk-engine.ts scoreReport() returns Verified Warning, Elevated Risk, or False Information — deterministic, 23 unit tests.
5	The system records a reason/evidence summary for the classification.	classifications.rationale_en (bilingual) plus evidence_snapshot (jsonb) and engine_version are stamped on every verdict.
6	A verified/elevated-risk result can be transformed into an actionable alert.	lib/core/alert-templates.ts builds a channel-specific, classification-keyed message.
7	Alerts can be sent through at least the configured pilot channels that are technically available.	alert_deliveries records per-channel status for WhatsApp, SMS, and the ambassador queue.
8	Youth ambassadors can receive/annotate relevant cases.	The ambassador PWA + ambassador_observations table, RLS-scoped to each ambassador's own pilot_area_id.
9	All key report, verification, classification and alert events are logged.	audit_events is append-only — no UPDATE/DELETE grant to any role.
10	Pilot metrics can be calculated from recorded events rather than manually typed demo values.	pilot_metrics is a SQL view computed live from reports, classifications, alerts, and ambassador_observations.
11	The system handles unavailable/conflicting data without falsely claiming certainty.	insufficient_evidence is a distinct boolean that can never silently collapse into False Information — enforced by dedicated unit tests.

#	Requirement	How the system satisfies it
12	An operator can review and override/escalate an uncertain or high-impact case.	The escalations table + Admin → Escalations page require a resolution note before a human can close a case.

Source: FloodFact AI Requirements Specification §18 — Acceptance Criteria for Phase 1 MVP.

6. Risk Register

Risk	Mitigation
Named data feeds (ArcGIS Living Atlas layer, KMD) not finalized before pilot start.	Open-Meteo (real, keyless) and a seeded PostGIS fallback already cover rainfall and flood-risk evidence, honestly labeled “sandbox” quality until real feeds are confirmed.
WhatsApp Cloud API's free test tier caps recipients before production verification completes.	SMS via Africa's Talking and the public web form remain fully functional fallback channels throughout.
Risk-engine miscalibration in the pilot's early days.	Week 2 shadow-mode human review of every classification before any goes live publicly; 23 unit tests already cover the documented safety rules.
Low household smartphone/data access limits WhatsApp reach.	SMS plus 3 physical community notice boards per ward; ambassadors as an in-person channel of last resort.
Ambassador attrition or unavailability during the 4-week window.	10 trained ambassadors per ward, not 1, gives built-in redundancy; training materials are reusable for fast onboarding of a replacement.
Short-rains onset timing shifts outside the planned 4-week window.	The system and ambassador workflow stay operational at low intake volume regardless; the pilot period can extend with no code change, since hazard type and pilot area are data, not code.

7. Cost & Scale

Every provider chosen for Phase 1 has a free or sandbox tier sufficient for 500 households. The architecture already generalizes — hazard_type and pilot_area are data, not code — so scaling is a provider-tier and message-volume question, not a rewrite. Dollar figures below are illustrative order-of-magnitude, pending final provider quotes at each tier.

Scale	Compute / database	Messaging	AI (LLM + embeddings)
500 households (Phase 1 pilot)	Vercel Hobby + Supabase Free — \$0	WhatsApp Cloud API free test tier + Africa's Talking sandbox — near \$0	Pay-per-token, low volume — low single-digit \$/month
5,000 households (Phase 2)	Vercel Pro + Supabase Pro — needed for backups and connection limits	WhatsApp/SMS production pricing becomes the largest line item	Same providers, ~10× token volume
50,000 households (Phase 3, national)	Supabase scaled tier / read replicas — same codebase, no rewrite	Messaging cost dominates total spend, not engineering	Batched extraction keeps marginal LLM cost per report low

8. Post-Pilot Roadmap

- **Phase 2** — Add landslide and fire hazards — new provider adapters and rows, same pipeline. No rewrite: hazard_type is data.
- **Phase 3** — Expand to SMS, radio, TV, and USSD channels at national scale across Nairobi's informal settlements.

Appendix — Companion Materials

- **Concept:** Problem, functionality, and innovation narrative — FloodFact-AI-Concept.pdf.
- **Demonstration:** Real screenshots of the running product, end to end — FloodFact-AI-Demonstration.pdf.
- **Repository:** Full source, migrations, tests, and docs/SETUP.md for going live — github.com/VinceOdundo/FloodfactAI.